Lean irreducibility of BINSTD degree-eight factors

Validation audit, 22 October 2026

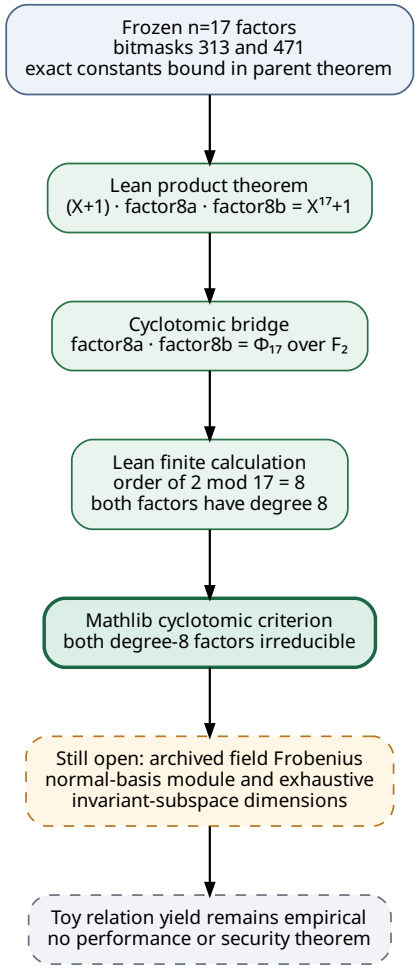
Frozen source. The n=17 BINSTD Stage-1 files list bitmask polynomials 313 and 471. The parent Lean task binds these exact coefficient patterns and proves that their product with $X + 1$ is $X^{17} + 1$ over $\mathbf{F}_2$.

New theorem. The two degree-eight polynomials

$$X^8 + X^5 + X^4 + X^3 + 1, \qquad X^8 + X^7 + X^6 + X^4 + X^2 + X + 1$$

are both irreducible in $(\mathbf{Z}/2\mathbf{Z})[X]$. The proof identifies their product with Φ_{17}, checks that 2 has multiplicative order 8 modulo 17 and that both factors have degree 8, then applies Mathlib's cyclotomic factor-degree criterion. The formal task targets their conjunction. The verification receipt reports a passing full build, standard axioms only across 625 audited theorems, and a passing target audit under Lean 4.35.0-rc1. Independent semantic review remains pending.

Boundary. This does not yet identify the archived field's Frobenius with the cyclic normal-basis module or classify all invariant subspaces. A separate open problem names that bridge. No empirical yield, performance or security claim follows.


Frozen n=17 factors
bitmasks 313 and 471
exact constants bound in parent theorem
Lean product theorem
(X+1) · factor8a · factor8b = X^17+1
Cyclotomic bridge
factor8a · factor8b = Φ17 over F2
Lean finite calculation
order of 2 mod 17 = 8
both factors have degree 8
Mathlib cyclotomic criterion
both degree-8 factors irreducible
Still open: archived field Frobenius
normal-basis module and exhaustive
invariant-subspace dimensions
Toy relation yield remains empirical
no performance or security theorem